Notes: Mian & Sufi (QJE, 2009)

The Consequences of Mortgage Credit Expansion: Evidence from the U.S. Mortgage Default Crisis

Contents

1 Big picture

- **Question.** What caused the rapid expansion of mortgage credit before the U.S. mortgage default crisis, and what explains why defaults were concentrated in subprime neighborhoods?
- **Core empirical design.** The paper uses ZIP-code level data and compares subprime and prime ZIP codes *within the same county*. This strips out much of the broad local housing market and labor market variation.
- **Subprime definition.** A ZIP code's subprime exposure is the fraction of residents with a credit score below 660 in 1996. Using 1996 avoids measuring subprime status after the credit boom itself may have changed credit scores.
- **Main fact 1.** Subprime ZIP codes experienced unusually strong mortgage credit growth from 2002 to 2005.
- **Main fact 2.** The same ZIP codes experienced sharply higher mortgage default growth from 2005 to 2007.
- **Main fact 3.** The mortgage credit boom was not supported by income growth. From 2002 to 2005, income growth and mortgage credit growth were negatively correlated across ZIP codes.
- **Historical contrast.** This negative income-credit relation is unusual: in other periods, income growth and mortgage credit growth are usually positively related.
- **Supply-side interpretation.** The evidence is consistent with an outward shift in the supply of mortgage credit to subprime borrowers.
- **Securitization channel.** ZIP codes with more subprime borrowers saw larger increases in mortgage sales to non-GSE investors, especially securitized pools and noncommercial bank financial firms.
- **Interest rate alternative.** Falling interest rates alone do not explain the pattern: an earlier falling-rate episode from 1990 to 1994 did not generate a comparable relative subprime mortgage boom.
- **House price alternative.** House price growth rose during the boom, but the paper argues house prices are more likely part of a feedback mechanism than an independent fundamental driver.
- **Economic takeaway.** The crisis was not just a story of local income growth or better fundamentals in poor neighborhoods. It was a credit supply expansion to high-risk borrowers, closely tied to securitization, that helped generate both mortgage growth and later defaults.

2 Data and measurement

2.1 Geographic unit and sample

- The main unit of observation is the ZIP code.
- The sample contains 3,014 ZIP codes in 166 counties, focused on the largest metropolitan areas.
- The paper emphasizes within-county variation because credit growth, default growth, and subprime borrower shares vary substantially inside counties.
- County fixed effects compare neighborhoods that share the same broad local economy and housing market.

Interpretation: The central empirical comparison is not Arizona versus Ohio or Miami versus Detroit. It is high-subprime versus low-subprime ZIP codes within the same county.

2.2 Credit quality and subprime share

- The key pre-boom risk measure is the fraction of residents with a credit score below 660 in 1996.
- The 660 cutoff is important because it was widely used by mortgage market participants to distinguish lower-risk borrowers from subprime borrowers.

- Prime ZIP codes are the lowest quartile of the national distribution of the subprime share.
- Subprime ZIP codes are the highest quartile of the same distribution.

Interpretation: This variable is predetermined relative to the 2002-2005 credit expansion. It is therefore useful for identifying where credit supply later expanded.

2.3 Credit, default, and mortgage origination data

- **Equifax:** ZIP-level mortgage debt, non-mortgage debt, mortgage default rates, and non-mortgage default rates.
- **HMDA:** mortgage originations, mortgage applications, denial rates, and secondary market sales of mortgages.
- **Fiserv Case-Shiller-Weiss and Zillow:** ZIP-level house price growth.
- **IRS:** adjusted gross income of residents living in each ZIP code.
- **Census and Census business statistics:** demographics, employment, establishments, education, poverty, homeownership, and race.
- **CAP Index:** ZIP-level crime statistics.

Interpretation: The empirical strategy depends on matching credit outcomes, income outcomes, house prices, and securitization behavior at a very fine geographic level.

2.4 Key outcome variables

The main outcomes are growth rates or changes, including:

$$\begin{aligned} \text{Growth}^{mortgage\ orig}_{zc,2002-2005}, &\quad \text{Growth}^{mortgage\ debt}_{zc,2002-2005}, \\ \text{Growth}^{income}_{zc,2002-2005}, &\quad \Delta\text{Default}^{mortgage}_{zc,2005-2007}, \\ \Delta\text{DeniedRate}_{zc,2002-2005}, &\quad \Delta\text{SoldToNonGSE}_{zc,2002-2005}. \end{aligned}$$

Interpretation: The paper distinguishes mortgage credit from non-mortgage credit. This matters because a broad household debt boom would show up in non-mortgage debt as well; the evidence is concentrated in mortgages.

2.5 Prime versus subprime ZIP codes

Subprime ZIP codes differ sharply from prime ZIP codes before the boom:

- subprime borrower fraction: 0.444 versus 0.159;
- fraction of FHA-backed loans: 0.239 versus 0.041;
- application denial rate: 0.307 versus 0.148;
- homeownership rate: 0.557 versus 0.812;
- median income: $38.9k versus $76.4k;
- poverty rate: 0.169 versus 0.038;
- nonwhite share: 0.418 versus 0.079.

Interpretation: These are high-credit-risk, lower-income neighborhoods. The key puzzle is why these neighborhoods received so much new mortgage credit just before defaults exploded.

3 Models and empirical specifications

3.1 Baseline within-county regression

The core specification is:

$$Y_{zc} = \alpha_c + \beta \text{SubprimeShare}_{z,1996} + \Gamma' X_{zc} + \varepsilon_{zc}, \tag{1}$$

where z indexes ZIP code and c indexes county.

- Y_{zc} is mortgage credit growth, income growth, default growth, denial-rate change, securitization growth, or house price growth.
- α_c are county fixed effects.
- $\text{SubprimeShare}_{z,1996}$ is the share of residents with credit score below 660 in 1996.
- X_{zc} includes controls such as income, employment, establishments, crime, and housing supply variables depending on the specification.

Interpretation: A positive β in a mortgage credit regression means subprime ZIP codes grow faster than prime ZIP codes in the same county. A negative β in an income regression means the same ZIP codes have weaker income growth.

3.2 Income-based hypothesis test

The income-based hypothesis predicts that credit growth should be explained by income, employment, or establishment growth:

$$\text{Growth}^{mortgage\ orig}_{zc,2002-2005} = \alpha_c + \beta \text{SubprimeShare}_{z,1996} + \gamma \text{Growth}^{income}_{zc,2002-2005} + \delta' Z_{zc} + u_{zc}. \tag{2}$$

Interpretation: If subprime credit expansion reflected improving fundamentals, subprime ZIP codes should have stronger income or employment growth. The paper finds the opposite.

3.3 Historical credit-income correlation

The paper estimates the correlation between income growth and mortgage origination growth over multiple historical windows:

$$\text{Growth}^{mortgage\ orig}_{zc,t_0,t_1} = \alpha_c + \theta_{t_0,t_1} \text{Growth}^{income}_{zc,t_0,t_1} + u_{zc,t_0,t_1}. \tag{3}$$

Interpretation: The key parameter is θ. In normal periods, it is positive. During 2002-2005, it becomes negative, which is evidence of an unusual decoupling of mortgage credit from borrower fundamentals.

3.4 Securitization channel

To test the supply-side/securitization mechanism, the paper estimates:

$$\Delta \text{SoldToInvestor}_{zc,2002-2005} = \alpha_c + \beta \text{SubprimeShare}_{z,1996} + \Gamma' X_{zc} + u_{zc}. \tag{4}$$

It also connects secondary-market sales to later defaults:

$$\Delta \text{Default}^{mortgage}_{zc,2005-2007} = \alpha_c + \lambda \Delta \text{SoldToInvestor}_{zc,2002-2005} + \Gamma' X_{zc} + u_{zc}. \tag{5}$$

Interpretation: The channel requires two links: subprime ZIP codes get more loans sold into private securitization channels, and those channels are associated with later default increases.

3.5 Interest rate alternative

The paper compares two falling-rate episodes:

- 1990-1994: Treasury yields fall sharply, but subprime mortgage origination does not surge relative to prime ZIP codes.
- 2001-2005: Treasury yields fall and subprime mortgage origination surges.

Interpretation: Falling rates are not sufficient. The distinct feature of the 2001-2005 episode is the rise of mortgage securitization and relaxed lending constraints.

3.6 House price alternative and feedback

The paper tests whether house prices can explain credit growth:

$$\text{Growth}_{zc,t_0,t_1}^{houseprice} = \alpha_c + \theta_{t_0,t_1} \text{Growth}_{zc,t_0,t_1}^{income} + u_{zc,t_0,t_1}. \tag{6}$$

Interpretation: If house prices reflect fundamentals, income growth should be positively correlated with house price growth. During the securitization years, this correlation turns negative, suggesting that house price growth may be partly caused by credit expansion rather than the reverse.

4 Identification and empirical design

4.1 What identifies the main effects

- Identification comes from cross-ZIP variation in 1996 subprime borrower shares within the same county.
- County fixed effects absorb county-wide economic conditions, house price trends, and credit market shocks.
- The key regressor is predetermined before the 2002-2005 credit expansion.
- Historical periods provide placebo-style comparisons for the income-credit relation.

Interpretation: The empirical design asks whether neighborhoods that were observably riskier before the boom received disproportionate new mortgage credit during the boom.

4.2 Why within-county comparisons matter

- Across metropolitan areas, credit growth can look correlated with income growth.
- Within metropolitan areas and counties, the boom is concentrated in lower-income, high-subprime neighborhoods.
- Using aggregate MSA-level data can therefore produce misleading conclusions.

Interpretation: Aggregation hides the core fact: the credit expansion was not simply flowing to cities with strong fundamentals; within cities, it flowed to weak-credit neighborhoods.

4.3 Why the 1996 subprime share is useful

- It predates the mortgage credit expansion.
- It captures borrower risk before securitization changed access to credit.
- It is measured broadly for residents, not only for loans labeled "subprime."

Interpretation: This avoids defining treatment based on loans that were themselves created during the boom.

4.4 Key robustness logic

- Non-mortgage debt does not expand in the same way as mortgage debt.
- Very small geographic fixed effects, including three-square-mile blocks, do not eliminate the result.
- High housing-supply elasticity MSAs still show subprime mortgage credit growth, securitization growth, and default growth.
- The 1990-1994 falling-rate period does not show comparable subprime mortgage expansion.

Interpretation: These checks push against explanations based only on local income, local housing supply, interest rates, or general household credit demand.

4.5 What the paper cannot fully prove

- Securitization is not randomly assigned across ZIP codes.
- House price growth and credit growth may interact through feedback loops.
- ZIP-level data cannot directly observe every lender's beliefs or borrower expectations.
- The analysis shows strong patterns and correlations, not a single randomized causal shock.

Interpretation: The paper is strongest as evidence that the dominant narrative of improving subprime borrower fundamentals is inconsistent with ZIP-level facts.

5 Tables and figures

The visuals are directly extracted from the paper. Adjacent tables and figures are grouped to save space and improve reading speed.

5.1 Figure I and Table I: aggregate correlations versus ZIP-level facts

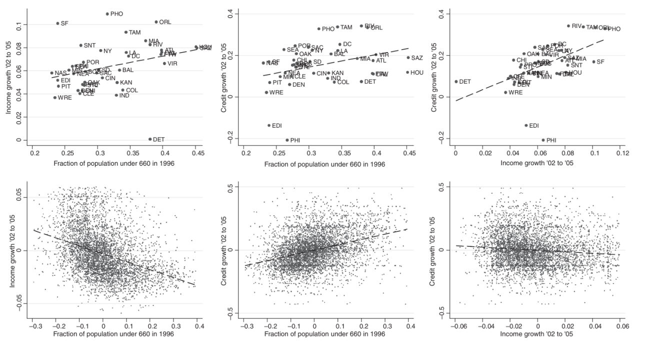

FIGURE I

Income Growth, Mortgage Credit Growth, and Subprime Population: Between- and Within-MSA Correlations

This figure presents correlations between income growth, mortgage credit growth, and the fraction of the population with a credi score under 660. The top panels present between-MSA correlations for the top forty MSAs by population, and the bottom panels utiliz ZIP code–level data to show the within-MSA correlations for the top forty MSAs. The bottom panels present data that are deviated fron MSA level means.

(a) Figure I: income, credit growth, and subprime population.

TABLE I

SUMMARY STATISTICS

	Mean	SD	Between-county SD	Within-county SD
Equifax data				
Mortgage debt annualized growth, 1996–2002	0.089	0.065	0.048	0.058
Non-home debt annualized growth, 1996–2002	0.073	0.092	0.040	0.087
Mortgage debt annualized growth, 2002–2005	0.145	0.081	0.049	0.071
Non-home debt annualized growth, 2002–2005	0.058	0.064	0.034	0.059
Mortgage default rate, 1996	0.030	0.024	0.013	0.021
Non-home default rate, 1996	0.068	0.035	0.019	0.030
Mortgage default rate change, 1996–2005	−0.003	0.028	0.018	0.023
Non-home default rate change, 1996–2005	−0.008	0.031	0.018	0.026
Mortgage default rate change, 2005–2007	0.035	0.038	0.024	0.032
Non-home default rate change, 2005–2007	0.018	0.026	0.012	0.024
Subprime consumer fraction (under 660), 1996	0.287	0.113	0.063	0.094
Fiserv Case–Shiller–Weiss data				
House price annualized growth, 1996–2002	0.096	0.033	0.034	0.011
House price annualized growth, 2002–2005	0.153	0.078	0.078	0.016
HMDA data				
Mortgage origination for home purchase ann. growth 1996–2002	0.144	0.089	0.068	0.067
Mortgage origination for home purchase ann. growth 2002–2005	0.194	0.189	0.130	0.150
Fraction sold to nonagency investors, 1996	0.253	0.081	0.078	0.045
Change in fraction sold, 1996–2002	0.044	0.057	0.045	0.043
Change in fraction sold, 2002–2005	0.241	0.051	0.038	0.038
IRS and Census statistics of U.S. business				
Income annualized growth, 1991–1998	0.052	0.025	0.012	0.022
Income annualized growth, 1998–2002	0.023	0.020	0.011	0.017
Income annualized growth, 2002–2005	0.048	0.035	0.021	0.031
Income annualized growth, 2005–2006	0.041	0.044	0.023	0.040
Employment annualized growth, 2002–2005	0.016	0.058	0.030	0.053
Establishment annualized growth, 2002–2005	0.017	0.029	0.019	0.024

(b) Table I: summary statistics.

Read:

- Across MSAs, credit growth appears positively correlated with income growth.
- Within MSAs, subprime neighborhoods have weaker income growth but stronger mortgage credit growth.
- Table I shows mortgage debt growth accelerates from 8.9% in 1996-2002 to 14.5% in 2002-2005, while non-home debt growth slows from 7.3% to 5.8%.
- Mortgage default rates rise by 3.5 percentage points from 2005 to 2007 after being roughly flat from 1996 to 2005.

Economic message: The aggregate picture can make the boom look fundamental. The ZIP-level picture shows that the boom flowed to neighborhoods with weaker income growth and higher pre-boom credit risk.

5.2 Table II and Figure II: subprime ZIP codes and the credit/default boom

TABLE II
SUBPRIME VERSUS PRIME ZIP CODES

	Prime ZIP codes	Subprime ZIP codes
Measures of mortgage credit availability		
Fraction of subprime borrowers in 1996 (under 660)	0.159	0.444**
Fraction of loans backed by FHA in 1996	0.041	0.239**
Fraction of mortgage applications denied, 1996	0.148	0.307**
Home ownership rate, 2000	0.812	0.557**
Demographic variables from Census 2000		
Median household income ($000)	76.4	38.9**
Poverty rate	0.038	0.169**
Fraction with less than high school education	0.077	0.289**
Fraction unemployed	0.031	0.081**
Fraction nonwhite	0.079	0.415**

Notes. This table presents characteristics of prime and subprime ZIP codes in our sample. Prime and subprime ZIP codes are determined by splitting ZIP codes into four quartiles based on the national distribution of the fraction of subprime borrowers (credit score less than 660) as of 1996. Prime ZIP codes are the lowest quartile and subprime ZIP codes are the highest quartile.

**,* Difference between prime and subprime statistically distinct from 0 at the 1% and 5% levels, respectively.

(a) Table II: subprime versus prime ZIP codes.

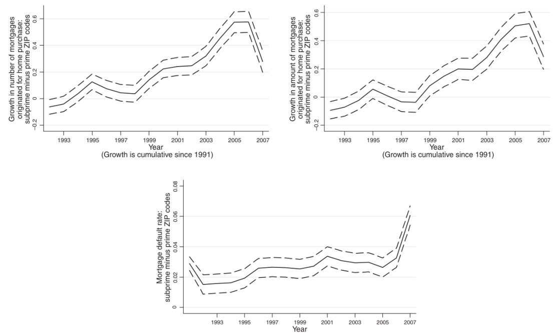

FIGURE II
Mortgage Credit Growth and Default Rates: Subprime Relative to Prime ZIP Codes

This figure plots the growth in the number (top left-hand panel) and amount (top right-hand panel) of originated mortgages and the mortgage default rate (bottom panel) for subprime relative to prime ZIP codes in the same county. Subprime and prime ZIP codes are defined to be the highest and lowest quartile ZIP codes in the national distribution based on the fraction of residents with a credit score less than 660 as of 1991.

(b) Figure II: mortgage credit growth and default rates.

Read:

- Subprime ZIP codes have much lower income, lower homeownership, higher unemployment, higher poverty, and more prior credit constraints than prime ZIP codes.
- Figure II shows subprime ZIP codes receive much faster mortgage origination growth from 2002 to 2005.
- The same ZIP codes experience a sharp increase in mortgage default rates after 2005.
- The default increase occurs after the credit expansion, not before it.

Economic message: The neighborhoods that were initially riskier became the center of mortgage lending growth and then of mortgage defaults.

5.3 Figure III and Table III: rejecting the income-based hypothesis

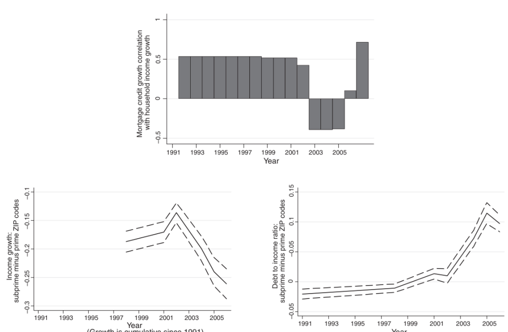

FIGURE III
Mortgage Credit Growth and Income Growth over Time

The top panel in this figure plots the correlation across ZIP codes between income growth and mortgage credit growth over after they are deviated from county means. The bottom left-hand panel plots the relative income growth for subprime relative to ZIP codes in the same counties, and the bottom right-hand panel plots the relative debt-to-income ratio for subprime relative to ZIP codes in the same counties. The debt-to-income ratio is defined as total originated mortgages for home purchase in a ZIP code by the total income of the ZIP code. Subprime and prime ZIP codes are defined to be the highest and lowest quartile ZIP codes national distribution based on the fraction of residents with a credit score less than 660 as of 1991.

(a) Figure III: mortgage credit and income growth over time.

TABLE III
CAN PRODUCTIVITY/INCOME GROWTH EXPLAIN SUBPRIME CREDIT EXPANSION FROM 2002 TO 2005?

	Mortgage origination growth 2002–2005 (1)	Income growth 2002–2005 (2)	Employment growth 2002–2005 (3)	Establishment growth 2002–2005 (4)
Fraction subprime borrowers, 1996	0.469**	−0.141**	−0.074**	−0.042**
	(0.029)	(0.006)	(0.011)	(0.005)
N	2,946	2,946	2,946	2,946
R^2	.42	.35	.15	.33

Notes. This table presents the correlation across ZIP codes between measures of income growth, employment growth, and business growth and the fraction of subprime borrowers. All regressions include county fixed effects.

**,*Statistically distinct from 0 at the 1% and 5% levels, respectively.

(b) Table III: can income or productivity explain credit growth?

Read:

- Table III shows that a higher 1996 subprime share predicts stronger 2002-2005 mortgage origination growth: coefficient 0.469.

- The same subprime share predicts weaker income growth: coefficient -0.141.
- It also predicts weaker employment growth and establishment growth.
- Figure III shows the credit-income correlation turns negative during the credit boom years.

Economic message: The mortgage boom is not explained by stronger local productivity or resident income growth in subprime neighborhoods.

5.4 Table IV and Table V: historical correlations and robustness

TABLE IV
HISTORICAL MORTGAGE CREDIT GROWTH AND INCOME GROWTH CORRELATIONS

	Dependent variable: Mortgage originations for home purchase growth, annualized							
	2002–2005 (1)	1991–1998 (2)	1998–2001 (3)	2001–2002 (4)	2002–2004 (5)	2004–2005 (6)	2005–2006 (7)	2006–2007 (8)
Income growth, annualized	−0.662**	0.537**	0.517**	0.425	−0.394**	−0.383**	0.103	0.716**
	(0.089)	(0.084)	(0.092)	(0.368)	(0.122)	(0.077)	(0.078)	(0.093)
N	3,014	2,809	3,014	3,014	3,014	3,014	3,014	3,014
R^2	.34	.55	.27	.44	.24	.39	.27	.26

Notes. This table presents the correlation across ZIP codes between mortgage origination for home purchase growth and income growth for different periods of our sample. Income growth and mortgage origination growth are measured for the exact same period for all specifications except the specification reported in column (7). We do not have income data available for 2007; as a result, in column (7) we examine the correlation between mortgage origination growth from 2006 to 2007 and income growth from 2005 to 2006. All specifications include county fixed effects.

**,*Coefficient estimate statistically distinct from 0 at the 1% and 5% levels, respectively.

Figure 4: Table IV: historical mortgage credit growth and income growth correlations.

Read:

- The mortgage credit-income coefficient is -0.662 in 2002-2005.
- In earlier periods, it is positive: 0.537 in 1991-1998 and 0.517 in 1998-2001.
- The sign turns back positive by 2006-2007: 0.716.

Economic message: 2002-2005 is historically unusual: mortgage credit grew fastest in ZIP codes with weaker income growth.

TABLE V
MORTGAGE CREDIT EXPANSION IN SUBPRIME ZIP CODES

	Annualized mortgage origination for home purchase growth, 2002–2005				Mortgage debt growth, 2002–2005	Non-mortgage debt growth, 2002–2005
				3-square-mile-block fixed effects		
	(1)	(2)	(3)	(4)	(5)	(6)
Fraction of subprime borrowers, 1996	0.460**	0.458**	0.454**	0.431**	0.050**	−0.039**
	(0.033)	(0.033)	(0.035)	(0.079)	(0.015)	(0.013)
Income growth, 2002–2005	−0.064	−0.068	−0.084	0.145	0.360**	0.201**
	(0.097)	(0.097)	(0.104)	(0.223)	(0.044)	(0.037)
Establishment growth, 2002–2005	0.084	0.096	0.118	−1.898**	0.805**	0.546**
	(0.124)	(0.127)	(0.142)	(0.314)	(0.058)	(0.049)
Employment growth, 2002–2005	−0.054	−0.053	−0.041	0.001	−0.040	0.001
	(0.056)	(0.056)	(0.056)	(0.101)	(0.026)	(0.021)
Crime growth, 2002–2005		−0.105	−0.012	0.551	0.251*	0.240*
		(0.256)	(0.278)	(0.779)	(0.117)	(0.098)
House price elasticity with respect to income			−0.014**	−0.010		
			(0.005)	(0.009)		
House quantity elasticity with respect to income			−0.001	0.014		
			(0.004)	(0.008)		
Fraction of housing units vacant, 2000			0.371**	0.865**		
			(0.074)	(0.252)		
Fraction of housing stock built last 2 years, 2000			0.363	1.161		

(a) Table V, first page.

TABLE V
(CONTINUED)

	Annualized mortgage origination for home purchase growth, 2002–2005				Mortgage debt growth, 2002–2005	Non-mortgage debt growth, 2002–2005
				3-square-mile-block fixed effects		
	(1)	(2)	(3)	(4)	(5)	(6)
Fraction of housing stock built 2–5 years ago, 2000			−0.108	0.162		
			(0.078)	(0.246)		
N	2,946	2,946	2,782	2,782	2,946	2,946
R^2	.42	.42	.45	.94	.33	.26

Notes. This table presents coefficient estimates from specifications relating the growth in mortgage originations for home purchase in a ZIP code from 2002 to 2005 to the fraction of subprime borrowers in 1996. Column (5) examines the growth in total mortgage debt, and column (6) examines the growth in non-home consumer debt. The measure of house price elasticity (house quantity elasticity) uses changes in median house value (number of owner-occupied housing units) from 1990 to 2000 from the decennial census, and changes in

(b) Table V, continuation.

Read:

- The subprime-share coefficient remains around 0.43-0.46 for home-purchase mortgage origination growth after adding controls.
- The coefficient is 0.050 for mortgage debt growth but -0.039 for non-mortgage debt growth.
- Very refined three-square-mile block fixed effects do not remove the effect.

Economic message: The credit expansion is robust, local, and mortgage-specific. It is not a general consumer credit boom or a simple housing-supply-control artifact.

5.5 Figure IV and Figure V: interest rates versus borrower constraints and securitization

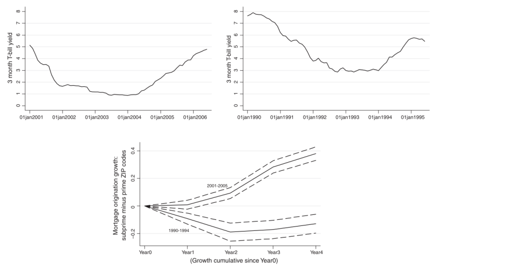

FIGURE IV
Relative Mortgage Origination Growth for Subprime ZIP Codes: Falling Interest Rate Periods

The top left- and right-hand panels show the evolution of the three-month Treasury bill yield during the five-year periods from 1990 to 1994 and 2001 to 2005, respectively. The bottom panel shows the growth in the amount of originated mortgages for subprime relative to prime ZIP codes in the same counties for these two five-year periods. Subprime and prime ZIP codes are defined to be the highest- and lowest-quartile ZIP codes in the national distribution based on the fraction of residents with a credit score below 660 as of the first year of the respective five-year periods.

(a) Figure IV: falling interest rate periods.

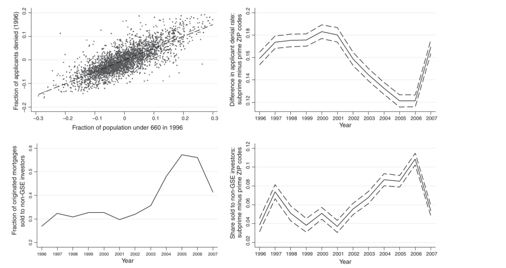

FIGURE V
Relaxation in Borrower Credit Constraints

The top left-hand panel shows the correlation across ZIP codes between the mortgage application denial rate and the fraction of residents with a credit score below 660, both as of 1996. The data are deviated from county means. The top right-hand panel shows the denial rate for mortgage applications for prime relative to subprime ZIP codes in the same county. The bottom left-hand panel shows the fraction of all originated mortgages for home purchase that are sold to non-GSE investors, and the bottom right-hand panel shows the relative fraction sold to non-GSE investors for subprime versus prime ZIP codes in the same county. Subprime and prime ZIP codes are defined to be the highest- and lowest-quartile ZIP codes in the national distribution based on the fraction of residents with a credit score below 660 as of 1996.

(b) Figure V: relaxation in borrower constraints.

Read:

- Figure IV compares 2001-2005 with 1990-1994, another period of falling Treasury yields.
- The relative subprime mortgage boom appears in 2001-2005 but not in the earlier falling-rate period.
- Figure V shows subprime ZIP codes had high denial rates before the boom, then experienced a large relative fall in denial rates.
- Figure V also shows a large rise in the fraction of mortgages sold to non-GSE investors in subprime ZIP codes.

Economic message: Falling rates are not enough. The boom coincides with relaxed borrower constraints and growth in private securitization channels.

5.6 Table VI and Figure VI: securitization and house prices

TABLE VI
EVIDENCE OF A SECURITIZATION CHANNEL

Panel A: Secondary mortgage sales and subprime ZIP codes

	Change in applicant denial rate 2002–05	Change during 2002–05 in the fraction sold to non-GSE investors	Change during 2002–2005 in the fraction sold to non-GSE investors who are			
			Affiliates	Commercial banks	Securitized pools Of mortgages	Noncommercial bank fin. firms
	(1)	(2)	(3)	(4)	(5)	(6)
Fraction of subprime borrowers, 1996	−0.094**	0.048**	−0.055**	−0.007*	0.104**	0.077**
	(0.006)	(0.009)	(0.005)	(0.003)	(0.004)	(0.004)
N	2,946	2,946	2,946	2,946	2,946	2,946
R^2	.58	.46	.56	.46	.68	.61

Panel B: Mortgage sales and changes in default rate

	Change in mortgage default rate from 2005 to 2007				
	(1)	(2)	(3)	(4)	(5)
Change during 2002–05 in the fraction sold to non-GSE investors	0.027				
	(0.015)				
Change during 2002–2005 in the fraction sold to non-GSE investors who are					
Affiliates		−0.247**			
		(0.027)			
Commercial banks			−0.116*		
			(0.046)		

(a) Table VI, first page.

TABLE VI
(CONTINUED)

	Change in mortgage default rate from 2005 to 2007				
	(1)	(2)	(3)	(4)	(5)
Securitized pools of mortgages				0.360**	
				(0.031)	
Noncommercial bank financial firms					0.314**
					(0.029)
N	2,946	2,946	2,946	2,946	2,946
R^2	.39	.40	.39	.41	.41

Notes. Panel A presents coefficient estimates relating the change in the fraction of originated mortgages sold in a ZIP code from 2002 to 2005 to the share of subprime borrowers as of 1996. Panel B presents estimates relating default rates from 2005 to 2007 to the fraction of loans sold by originators to investors from 2002 to 2005. All specifications include county fixed effects and control variables for income, wage, employment, business establishments, and crime growth.

**,*Coefficient estimate statistically distinct from 0 at the 1% and 5% levels, respectively.

(b) Table VI, continuation.

Read:

- Panel A shows that subprime ZIP codes experience a fall in application denial rates: coefficient -0.094.
- They also experience a rise in the fraction of originations sold to non-GSE investors: coefficient 0.048.

- The rise is especially strong for securitized pools of mortgages and noncommercial bank financial firms.
- Panel B connects these private securitization channels to later default growth.

Economic message: The evidence points toward private securitization and relaxed screening as mechanisms behind the subprime credit expansion.

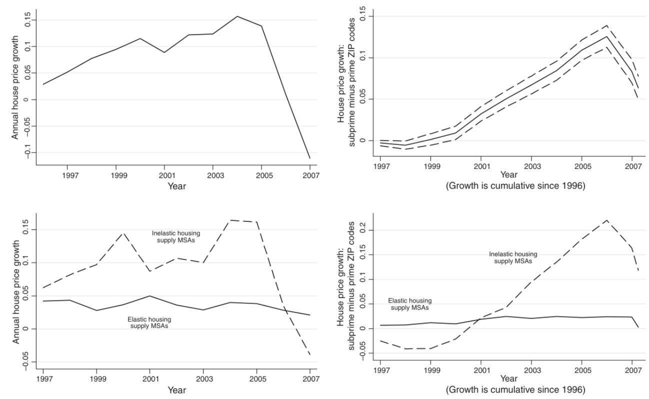

Figure 8: Figure VI: house price growth.

Read:

- House price growth rises during the boom and then falls sharply around the crisis.
- Subprime ZIP codes experience higher relative house price growth.
- The paper separates elastic and inelastic housing supply MSAs to test whether local housing supply explains the mortgage boom.

Economic message: House prices are important, but they are not cleanly independent fundamentals. The evidence suggests feedback between mortgage credit expansion and house price growth.

5.7 Table VII, Table VIII, and Figure VII: elastic MSAs and historical house price evidence

TABLE VII
MORTGAGE ORIGINATION GROWTH AND MORTGAGE DEFAULT RATE CHANGES FOR HIGH–HOUSING SUPPLY ELASTICITY MSAS

	Income growth 2002–2005	Change in fraction sold in securitizations 2002–2005	Change in fraction to other financial firms 2002–2005	Mortgage origination growth 2002–2005		Change in mortgage default rate 2002–2005	
					With controls		With controls
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Fraction subprime borrowers, 1996	−0.069**	0.100**	0.061**	0.305**	0.413**	0.057**	0.056**
	(0.010)	(0.009)	(0.014)	(0.061)	(0.069)	(0.015)	(0.018)
N	655	655	655	655	655	655	655
R^2	.17	.28	.43	.10	.12	.04	.05

Notes. This table examines relative income growth, securitization patterns, and mortgage origination growth from 2002 to 2005 and the relative change in mortgage default rates from 2005 to 2007 in high–subprime borrower share ZIP codes. The sample is limited to the top–decile MSAs by housing supply elasticity as measured by Saiz (2008). The ten high–housing supply elasticity MSAs are Dayton, OH; Fort Wayne, IN; Greenville, SC; Indianapolis, IN; Kansas City, MO; Little Rock, AR; McAllen, TX; Omaha, NE; Tulsa, OK; and Wichita, KS. The control variables in columns (5) and (7) are income growth, employment growth, and business establishment growth from 2002 to 2005. All growth rates are annualized, and all specifications include MSA fixed effects.
**,* Coefficient estimate statistically distinct from 0 at the 1% and 5% levels, respectively.

(a) Table VII: high-housing-supply-elasticity MSAs.

TABLE VIII
HISTORICAL HOUSE PRICE GROWTH AND INCOME GROWTH CORRELATIONS

	Dependent variable: House price growth, annualized							
	2002–2005	1991–1998	1998–2001	2001–2002	2002–2004	2004–2005	2005–2006	2006–2007
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Income growth, annualized	−0.118**	0.138**	0.070**	0.130**	−0.074**	−0.082**	−0.024*	0.195**
	(0.010)	(0.008)	(0.013)	(0.011)	(0.010)	(0.008)	(0.010)	(0.017)
N	3,014	2,875	3,014	3,014	3,014	3,014	3,014	2,721
R^2	.96	.90	.87	.89	.95	.95	.78	.87

Notes. This table presents the correlation across ZIP codes between house price growth and income growth for different periods of our sample. Income growth and house price growth are measured for the exact same period for all specifications except the specification reported in column (7). We do not have income data available for 2007; as a result, in column (7) we examine the correlation between house price growth from 2006 to 2007 and income growth from 2005 to 2006. All growth rates are annualized, and all specifications include county fixed effects.
**,* Coefficient estimate statistically distinct from 0 at the 1% and 5% levels, respectively.

(b) Table VIII: house price-income correlations.

Read:

- Table VII shows the key patterns survive in high-elasticity housing supply MSAs: subprime share predicts securitization, mortgage origination growth, and default growth.
- Table VIII shows that house price growth and income growth are negatively correlated from 2002 to 2005: coefficient -0.118.

- In earlier periods the relation is positive, such as 0.138 in 1991-1998 and 0.130 in 2001-2002.

Economic message: Even in elastic housing markets, the subprime credit boom is visible. The house price-income relationship during the boom is historically abnormal.

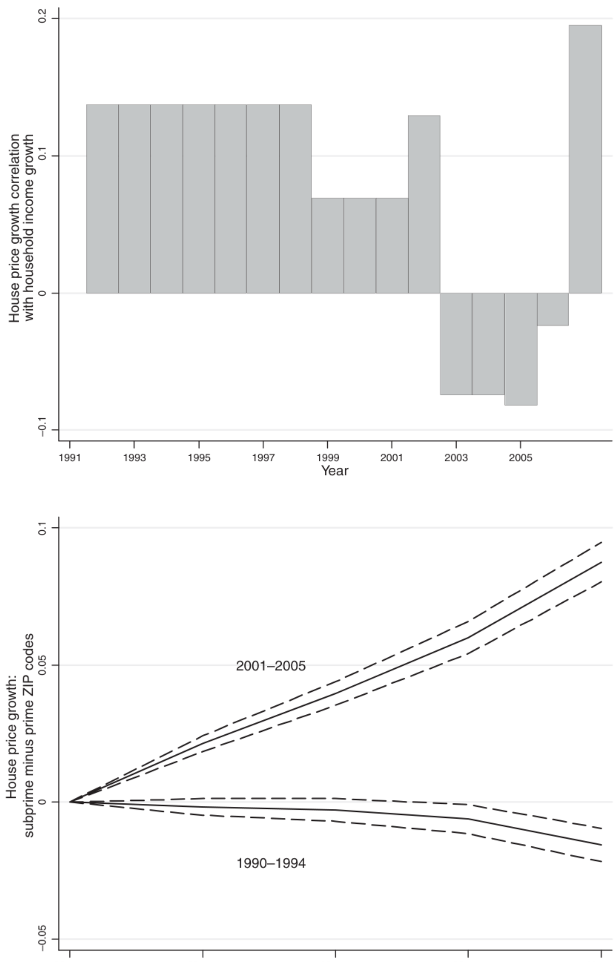

Figure 10: Figure VII: house price growth, historical evidence.

Read:

- The top panel shows the house price-income growth correlation turns negative during the securitization years.
- The bottom panel shows subprime ZIP codes outperform prime ZIP codes in house price growth during 2001-2005, but not during 1990-1994.

Economic message: The house price boom itself appears connected to the mortgage credit expansion. Treating house price growth as an exogenous fundamental may be misleading.

6 Short summary

- The paper studies the origins of the U.S. mortgage default crisis using ZIP-code level data.
- Subprime ZIP codes are defined using the fraction of residents with credit scores below 660 in 1996.

- Subprime ZIP codes have weaker income, higher poverty, lower homeownership, and higher denial rates before the boom.
- From 2002 to 2005, these ZIP codes receive disproportionate mortgage credit growth.
- From 2005 to 2007, these same ZIP codes experience a sharp rise in mortgage defaults.
- The mortgage credit boom is not matched by income, employment, or establishment growth.
- The credit-income correlation turns negative during 2002-2005, unlike most historical periods.
- The effect is mortgage-specific and not mirrored in non-mortgage consumer debt.
- Falling interest rates do not explain the pattern because the earlier 1990-1994 falling-rate period does not produce a similar boom.
- Securitization is closely linked to the credit expansion: subprime ZIP codes see a rise in loans sold to non-GSE investors.
- Private securitization channels are also associated with later default increases.
- House price growth is part of the story, but the paper treats it as potentially endogenous to credit expansion rather than as a clean fundamental driver.
- The broad conclusion is that the mortgage crisis was driven by a credit supply expansion to high-risk borrowers, closely tied to securitization.

7 Conclusion

7.1 Core conclusion

Mian and Sufi show that the pre-crisis mortgage boom was concentrated in high-subprime-share ZIP codes despite weak income growth. This pattern is historically unusual and inconsistent with a simple story in which credit expanded because subprime borrowers became more creditworthy.

7.2 Economic intuition

The intuition is that securitization changed the supply of mortgage credit. Lenders and originators were able to sell loans to non-GSE investors, especially private securitized pools and noncommercial bank financial firms. This relaxed borrowing constraints in neighborhoods that were previously credit-rationed. The immediate effect was mortgage credit growth; the later effect was a sharp rise in defaults.

7.3 Takeaway

The paper's central lesson is that financial innovation and credit supply can transform local credit allocation. Aggregate data make the boom look consistent with fundamentals. ZIP-level data reveal that the boom was directed toward high-risk neighborhoods with weak income growth, helping explain both the expansion and the subsequent default crisis.